Applied Physics B (2022) 128:137
https://doi.org/10.1007/s00340-022-07860-5

Applied Physics B
Lasers and Optics

Dynamic flat-topped laser beam shaping method using mixed region amplitude freedom algorithm

Dina Yaqoob Alsaka[1] · Çağlar Arpali[2] · Serap Altay Arpali[3] · Mohammed Fawzi Altemimi[1]

Received: 11 March 2022 / Accepted: 7 June 2022 / Published online: 2 July 2022
© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2022

Abstract
A dynamic beam shaping method is proposed for the generation of flat-top beams (FTBs) in the far field. Using the mixed-region amplitude freedom algorithm, this new method is used to design the required phase distribution encoded on a spatial light modulator for the generation of FTB profiles. The characteristics of these new beam shaping methods are used as beam parameters, such as the laser beam size, the beam intensity of square FTBs, and the root-mean-square error (RMSE). Using our proposed method, the theoretical performance of beam intensity shaping is improved to an RMSE < 0.02 with a minimum number of iterations of phase reconstruction. Using the phase hologram of dynamic beam shaping, theoretical and experimental comparisons of edge steepness and plateau uniformity were established for the square FTBs of variable beam sizes. It is shown that the dynamic beam shaping of FTBs can produce high intensity uniformity in the plateau region with steep edges, which makes it an effective tool, especially for laser machining applications.

1 Introduction

Laser beam shaping is a process of redistribution of the intensity and phase of electromagnetic radiation [1]. It has been widely used in fields such as material processing, lithography, medical applications, laser printing, optical data storage, micromachining in the electronics industry, isotope separation, optical processing, and laboratory research. These applications require a uniform beam that has a flattened irradiance profile with a uniform central region, and that can propagate through the working volume with limited distortion of its uniformity [2, 3]. There are many techniques for lossless transformation of Gaussian beams to uniform irradiance. Both refractive and reflective optical systems have been used to shape laser beam intensity profiles. The first paper on the shaping of a laser beam to uniform irradiance was published in 1965 [4]. The conversion of the Gaussian beam profile into a uniform profile has been achieved using a plano-convex lens [5], the polarization effect [6], a phase filter [7], two binary filters [8], multifaceted mirrors [9], spatial filters [10], deformable mirrors [11], nonaxisymmetric surface mirrors [12], negative aspherical aberrations [13], single aspherical lenses [14], and gradient index laser beam shapers [15]. The refractive and reflective techniques are limited in that they are tailored for a specific laser beam and its output pattern. There are several new techniques based on the principle of interference or diffraction effects that shape the irradiance profile of laser beam shaping systems. The components needed to effect diffraction phenomena on beam shaping are called diffractive optical elements (DOEs). DOEs can be used in a holographic approach using the inherent Fourier transform in the far field of the hologram for coherent illumination. Several different algorithms exist to transform a known incident spatial intensity distribution toward an FTB in the far field based on

✉ Çağlar Arpali
c.arpali@cankaya.edu.tr

Dina Yaqoob Alsaka
Dina.Y.Alsaka@nahrainuniv.edu.iq

Serap Altay Arpali
serapaltay.arpali@ostimteknik.edu.tr

Mohammed Fawzi Altemimi
Mohammed.Al-temimi@nahrainuniv.edu.iq

1 Department of Laser and Optoelectronics Engineering, Al-Nahrain University, Al Jadriyah Bridge, Baghdad 64074, Iraq

2 Department of Mechatronics Engineering, Çankaya University, Yukarıyurtçu Mahallesi Eskişehir Yolu 29. Km, Mimar Sinan Caddesi No: 4, 06790 Etimesgut/Ankara, Turkey

3 Department of Computer Engineering, Ostim Technical University, Ostim Osb/Yenimahalle, 100.Yıl Blv. 55/F, 06374 Ankara, Turkey

a DOE [16]. For example, a rotational symmetrical Gaussian beam is converted into a nearly FTB numerically using a single-phase DOE with an ST iterative algorithm [17]. In a comparison between the iterative Fourier transform algorithm (IFTA) and vector angular spectrum on adding the influence of the variation of the beam's polarization a super-Gaussian FTB was generated [18]. A phase plate is inserted in front of a lens to act as a DOE that converts the input Gaussian beam distribution into a top-hat beam [19]. The disadvantages of DOEs are static, and even a minor modification requires a complete change in the fundamental design, adding significant cost. An alternative, and much more flexible approach, is to use a liquid crystal spatial light modulator (SLM). In an SLM, each pixel of the device can produce a variable and controlled phase delay; hence, the SLM can be used as a variable DOE or it can be called a type of adaptive optics (AO) technology. Digitally generated phase patterns are displayed on a liquid crystal display panel controlled by a computer to dynamically generate custom phase patterns [20]. The development of the phase-only SLM device began when liquid crystals were discovered. Phase-only SLMs made it possible to create a wide range of novel laser beams, most of which cannot be created in any other way. They are based on a computer-generated hologram (CGH) and an IFTA that predicts the propagation of a beam through an initial kinoform by fast Fourier transform. It then successively modifies the kinoform based on a comparison between the predicted and desired Fourier plane intensities [21, 22]. Many researchers have used different algorithms to reshape laser beam profiles. For instance, annular, square, ring, circular, and super-Gaussian profiles of FTBs are generated depending on the phase patterns that are digitally generated [23, 24]. For dynamic laser beam shaping, a modified version of IFTAs and phase-only SLMs are used to create super-Lorentz and annular FTBs with variable diameters [21, 25]. The real-space implementation of CGH-based flat-topped beam shaping methods is proposed with the highest accuracy on sharp edges using the offset mixed-region amplitude freedom (MRAF) algorithm [26]. To increase the efficiency of beam shaping, new experiments were performed using a binary-amplitude SLM with an error diffusion algorithm [27]. Because the SLM has inter-pixel cross talks that degrade the phase modulation efficiency, different methods have been proposed to reduce speckle noise for flat-top beam generation. These include a modified Gerchberg–Saxton (GS) algorithm [28] or using time averaging of the constructed image of a multiple shifted hologram [29]. Other methods of speckle suppression at the detector plane include applying phase constraints during hologram calculations [30] or using two different SLMs [31]. Currently, based on all-optical neural network a new diffractive beam shaping method has been proposed to improve the beam shaping capacity with respect to ITFA [32]. Beam intensity profiling with a lower RMSE, is another important issue for high-precision applications that need to be addressed [27, 28]. Although the aforementioned methods improve the reconstruction of FTBs, they still suffer from speckle noise and increase the iteration time taken to reconstruct the beams in the far field with a low RMSE. To overcome these drawbacks, we propose an alternative beam shaping method that makes use of the dynamic control of the processing area of the MRAF algorithm of the IFTA. In the current study, we used the MRAF algorithm of the IFTA to generate controllable beam diameters of square FTBs in the far field. These square FTBs are generated by controlling the number of pixels inside the non-zero intensity area of the MRAF algorithm without changing the full number of pixels of the entire desired intensity distribution that is inserted in the Fourier plane. We call this new beam intensity controlling technique the modified MRAF (MMRAF) method. For the proof of concept, we introduced numerical and experimental comparisons between dynamically generated FTBs in terms of laser beam parameters such as beam size and beam intensity based on the MMRAF algorithm. In addition, to evaluate the flat-top spatial energy distributions of the generated square FTBs, we performed edge steepness and beam uniformity tests based on standard procedures and definitions [33, 34]. The competencies of these new beam shaping methods with respect to the classical MRAF algorithm were also used based on RMSE comparison.

Section 2 of this paper describes the equation of the generated FTBs relying on the kinoform hologram at the Fourier plane. It also presents the effect of the SLM on the FTBs using the MMRAF method. It then goes on to introduce the standard procedure and definition for the evaluation of the FTB parameters in the profile of the FTBs. The simulation test and theoretical and experimental results of square FTBs are presented in Sects. 3 and 4, respectively, with concluding remarks in Sect. 5.

2 Theory

2.1 Generation of dynamic FTBs using kinoform hologram

Depending on the Fraunhofer diffraction phenomena of light using an SLM and a computer-generated phase hologram, a new method is used to generate FTBs with controllable beam sizes and intensities. The relation between the incident beam profile in the input plane (SLM plane) and the output plane (Fourier plane or detector plane) is described using a Fourier transfer function [35]. To design and obtain the phase hologram of any desired light distribution, the kinoform is constructed by propagating the light field to and from the input plane (SLM plane) and the output plane

(Fourier plane) iteratively. Thus, in the process of generating these new FTBs, the MRAF is used as an IFTA. The most basic form of IFTA is the GS algorithm [29, 30]. The GS algorithm is based on the convergence of the phase at the SLM plane by applying constraints to the amplitude of the complex amplitudes and allowing the phase in both the SLM and Fourier planes to vary [36–38]. The set of equations based on the Fraunhofer diffraction integrals describes the iteration of the GS algorithm as:

$$U_{fp(k)}(x,y;z) = \mathcal{F}\{U_{in}(x',y';0)\} = \mathcal{F}\{U_i(x',y',0)exp[-i\emptyset_{\text{initial phase}}(x',y',0)]\}, \quad (1)$$

$$U_{fp'(k)}(x,y;z) = A(x,y;z)exp[-i\theta_p[U_{fp(k)}(x,y;z)]], \quad (2)$$

$$U_{in'(k)}(x',y',0) = \mathcal{F}^{-1}\{U_{fp'(k)}(x,y;z)\}, \quad (3)$$

$$U_{inc(k+1)}(x',y';0) = U_i(x',y';0)exp[-i\theta(U_{in'(k)}(x',y';0))], \quad (4)$$

where $\mathcal{F}$ is the Fourier operator;$\mathcal{F}^{-1}$ is the Fourier inverse operator; (x', y'), and (x, y), are the coordinate axes in the input and output planes, respectively; z is the distance between the input and output planes; k is the *kth* iteration;$U_{fp(k)}(x,y;z)$ defines the complex propagated field; $U_{in}(x',y';0)$ is the complex field with an initial phase at the SLM plane; $\varnothing_{\text{initialphase}}$ is the initial random phase added to the initial complex field;$A(x,y;z)$ is the desired amplitude distribution; θ_p is the phase of the reconstructed field at the diffraction plane; $U_{in'(k)}(x',y',0)$ defines the amplitude of the new complex field at the input plane with the GS algorithm; $U_{inc(k+1)}(x',y';0)$ is the iteratively generated new complex field at the input plane; $U_{fp'(k)}(x,y;z)$ is the enforced complex field at the diffraction plane; and θ is the phase kinoform hologram that is combined with the transfer functions of the digital focusing and blazed grating to obtain the final hologram. Thus, the final hologram encoded by the SLM [39] was:

$$\phi_{\text{SLM}} = \text{mod}\left[\frac{k}{2f}(x'^2 + y'^2) + 2\pi(G_x x' + G_y y') + \theta\right]. \quad (5)$$

where, G_x and G_y are the spatial frequencies of the grating in the x' and y' directions, k is the wave number, f is the focal length of the lens, and *mod* is the 2π modulo operation that creates the blazed grating [40]. Note that the convergence of the GS algorithm is extremely erratic because of the constraints on the amplitudes in the process planes. Thus, the MRAF algorithm improves the GS algorithm to overcome these constraints.

2.2 MMRAF method

The MRAF algorithm is an improved version of the GS algorithm, which is effective in solving the general amplitude-phase-retrieval problem by relaxing the restrictions on the amplitudes in the Fourier plane. This is achieved by partitioning the diffraction or Fourier plane into two parts: a signal region (*SR*) and a noise region (*NR*). The *SR* can be defined as all points of the desired intensity distribution with non-zero values, and the *NR* refers to the zero potential in the desired distribution [35]. Dividing the Fourier plane into subsets and determining the initial phase that is added to the initial complex field at the SLM plane, results in a neat convergence in the algorithm to the desired profile within the *SR*. It also reduces the dead space zones that cause the appearance of unmodulated light (dc term) in the reconstructed patterns when compared with the GS algorithm, while behavior in the *NR* falls under less control. During the iteration, the amplitude of the propagated field is replaced by the amplitude of the desired distribution inserted into the Fourier plane only for the pixels in the *SR*. This means that the usable area of the Fourier plane is reduced compared to that of the GS algorithm. Accordingly, Eq. (2) becomes, in terms of the MRAF algorithm:

$$U_{fp'mix}(x,y;z) = \left[m\left(|A(x,y;z)|^2\right)_{SR} + (1-m)\left(\left|A_p(x,y;z)\right|^2\right)_{NR}\right]exp[-i\theta_p(x,y;z)] \quad (6)$$

where m is the mixing parameter, which is responsible for the relative distribution of the optical power in the two regions. Its value is quantified as $0 \le m \le 1$, and $|\,.\,|$ is the absolute value. Equation (6) is called the mixed complex field at the Fourier plane generated from the combination of the desired and propagated fields, A andA_p, respectively. Therefore, the insertion of the desired amplitude pattern in Eq. (6) is a major step to set the diffracted field intensity closer to the desired pattern for each iteration [41]. From Eq. (6), the kinoform hologram is obtained by taking its argument, and applying it to Eq. (5) to obtain the final hologram encoded by the SLM device. Based on the division of the Fourier plane with the MRAF algorithm, we introduce a new method (MMRAF) to control the number of pixels inside the *SR*. This is called the controlled signal region (*SR′*). It is used to generate dynamic FTBs with different widths across the beam size of the original beam without changing the total number of pixels and describes the

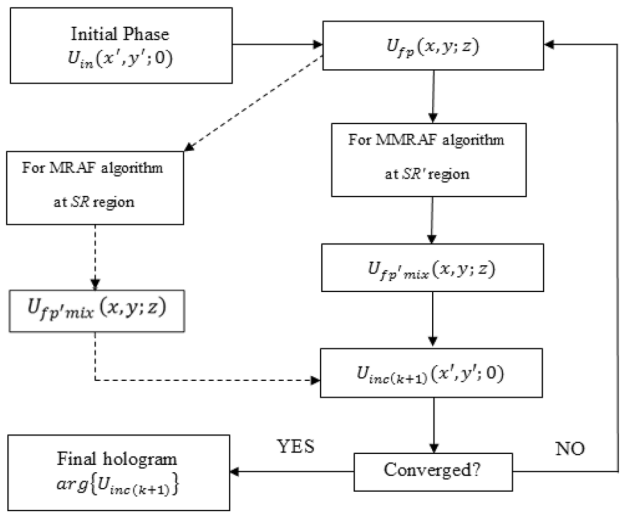

Fig. 1 Block diagram of MRAF and MMRAF algorithms

entire image of the desired intensity distribution. Thus in Eq. (6) each *SR* become *SR′*. Difference between MRAF and MMRAF algorithms is shown in Fig. 1 as block diagram representation. Therefore, the equation to describe the desired FTB field profile is based on the *SR′* of the MMRAF algorithm, which is assumed to be symmetrical in the *x* and *y* directions and is given by

$$U_{FTBMRAF}(x,y) = \left\{ \begin{array}{ll} 1 & m \leq \propto_x \leq n,\ m \leq \propto_y \leq n \\ 0 & otherwise \end{array} \right\}, \tag{7}$$

where $\propto_x$ and $\propto_y$ are the beam sizes (half width half maximum of the field) of the generated beam at the Fourier plane, which is assumed to be symmetrical in the *x* and *y* directions, respectively; *m* and *n* are the number of pixels defining the width of the *SR′* in the (– *x*, *x)* and (-*y*, *y*) directions at the Fourier plane, and are assumed to be equal in both directions. It should be noted that our results in this study are obtained for symmetric cases, but our method can also be applied to asymmetric flat beams. The intensity profile of the desired FTB [42] can be found as:

$$I_{FTB}(x,y) = \left|U_{FTBMRAF}(x,y)\right|^2 \tag{8}$$

Because the beam sizes are equal in both directions, we can name it as $\propto_{x,y}$ and it is a function of the *SR′* and the diffraction-limited pixel size at the Fourier plane, which is obtained from

$$\propto_{x,y} = \frac{SR'\Delta}{2} \tag{9}$$

where Δ is the diffraction-limited pixel size in the Fourier plane, which is computed by

$$\Delta = \frac{\lambda f}{N\Delta'} \tag{10}$$

where λ is the wavelength of the incident laser beam, f is the focal length, N is the maximum number of pixels of the SLM device, and Δ' is the SLM pixel pitch. From Eqs. (9) and (10), we notice that the diffraction-limited pixel size Δ is restricted by λ, f, N, and Δ', so it is a constant value, while *SR′* is a variable parameter with the number of pixels describing the full width of the FTB. Therefore, we can use Eqs. (7) to (9) as controllable parameters to generate different dynamic profiles of the FTB with different beam sizes and powers based on the phase kinoform hologram calculated by the MRAF algorithm.

2.3 Determination of initial phase to generate dynamic FTBs

The advantage of the MRAF algorithm in addition to dividing the Fourier plane into an *SR* and *NR* is the selection of the initial phase that is added to the initial beam profile. This is an extremely important issue for obtaining a perfect match between the desired intensity distribution and the intensity pattern of the initial beam profile. This selection is an effect of the degree of freedom on the convergence of the MRAF algorithm by reducing the number of iterations and the time required to obtain the diffraction patterns [40, 43].

In the current study, we propose a new methodology to calculate the initial phase. This is in preference to guessing the initial phase based on the controllable number of pixels used to generate the full width of the FTB and the diffraction-limited pixel size Δ. It is combined with a spot of light in the Fourier plane to conform to the desired full width of the FTB. The generation of a light spot at the output plane that approximately corresponds to the desired FTB takes the form:

$$\emptyset_1(x',y') = (SR'\Delta)(x'+y') \tag{11}$$

Equation (11) produces a spot with a controlled beam size depending on the width of the FTB (*SR′*) and the diffraction-limited pixel size (Δ). In addition, a linear gradient is in the form of:

$$\emptyset_2(x',y') = C(x'^2 + y'^2) \tag{12}$$

Equation (12) is used to move the envelope in these coordinates, relying on the value of parameter *C*.

Thus, we can write the equation for determining the initial phase as a combination of Eqs. (11) and (12) as:

$$\emptyset_{\text{cal}}(x', y') = (SR'\Delta)(x' + y') + C(x'^2 + y'^2). \tag{13}$$

In Eq. (13), the size of the created spot depends on the parameter $\boldsymbol{C}$, and the calculated initial phase as a function of the full width of the FTB, which is a function of $\propto_{x,y}$ and (x', y') are the coordinate axes in the SLM plane (input plane). The initial phase is assumed to obtain dynamic FTBs using the MMRAF algorithm. Equation (13) is added to the incident field at the SLM plane as an initial phase required to start the algorithm.

2.4 FTB pattern shaping using the SLM

Phase-only SLMs are simply arrays of rectangular apertures that are separated by a transparent material [44, 45]. They each comprise $Nx' \times Ny'$ pixels with an aperture of $a_{x'}, a_{y'}$ that are arranged at a pixel pitch of $\rho_{x'}$ and $\rho_{y'}$, respectively. Within the aperture area, the light can be modulated while the region between the pixels is inactive, and is usually masked for a transmissive SLM [45]. Phase-only SLMs change the phase of the incident light beam on each pixel. If adjacent pixels are made to oscillate in phase about two values, the required modulation can be achieved. Therefore, the transmittance function $t_{SLM}(x', y', 0)$ of an ideal SLM is shown in Eq. (14) and is the overall SLM aperture multiplied by the discrete complex amplitude of the pixel matrix [46]:

$$t_{\text{SLM}}(x', y', 0) = \text{rect}\left(\frac{x'}{L_{x'}}, \frac{y'}{L_{y'}}\right)\left\{\text{rect}\left(\frac{x'}{a_{x'}}, \frac{y'}{a_{y'}}\right) \otimes \left[\text{comb}\left(\frac{x'}{\rho_{x'}}, \frac{y'}{\rho_{y'}}\right)\exp\left[i\theta(x', y';0)\right]\right]\right\} \tag{14}$$

where $\otimes$ denotes a convolution operator; $\rho_{x'} = \rho_{y'} = \Delta'$ is the SLM pixel pitch in the x' and y' directions, respectively; $L_{x'}$ and $L_{y'}$ are the width and the length of the SLM, respectively; and θ is the kinoform hologram computed using the IFTAs. The complex field at the Fourier plane ($D = f$) is given by the exact Fourier transform of the complex field at the SLM plane, and the axes of the coordinates used in the Fourier transform analysis are illustrated in Fig. 2 [37, 46].

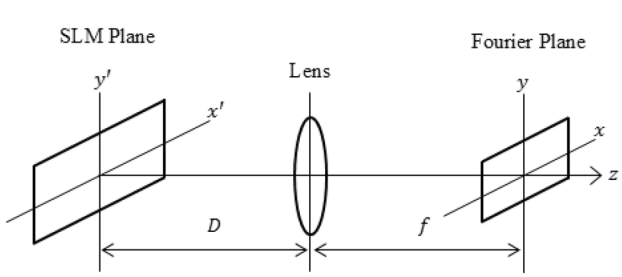

Fig. 2 Set of coordinate axes used in the Fourier transform analysis

$$U_{fp}(x, y;z) = \mathcal{F}\{U_{in}(x', y';0)t_{SLM}(x', y', 0)\}. \tag{15}$$

where $U_{in}(x', y';0)$ is the complex field with an initial phase at the SLM plane. This is given by [47]:

$$U_{in}(x', y';0) = U_i(x', y';0)\exp[-i\emptyset(x', y';0)]. \tag{16}$$

where $U_i(x', y';0)$ is the initial Gaussian distribution, which is expanded as a plane wave incident on the SLM plane. For the GS algorithm, $\varnothing(x', y';0)$ is assumed to be a random phase for the first iteration, whereas for the MMRAF method, it is calculated using Eq. (13). Therefore, the initial Gaussian distribution does not contain the phase information of the desired profile of the FTB. This leads to poor phase modulation, which is insufficient to attain exact beam shaping. Moreover, in execution, a greater number of iterations is required to obtain the information for the required shape. The phase in the next iteration $\varnothing_{SLM}^{(k+1)}$ is defined as the argument of the inverse Fourier transform of Eq. (6) for the MMRAF algorithm. The algorithm is recursive for the kth iteration to match the FTB distribution at the output plane. Equation (6) is used to obtain the reconstructed patterns of the FTB intensity distribution and the kinoform hologram θ, as shown in Eq. (14), which was calculated using the IFTA and added to Eq. (5) to yield the final kinoform hologram that loads to the SLM as the CGH of the dynamic beam shaping.

In this study, MATLAB (version R2013a) software was used as an effective program to calculate the final kinoform hologram θ from the MMRAF method to generate square FTBs.

2.5 FTB parameters

2.5.1 Edge steepness and plateau uniformity of the dynamic FTBs

In most cases, it is necessary to use flat-top spatial energy distributions with steep edges and high uniformity in the plateau region [2, 33]. Reliable standard methods for the evaluation of nearly FTB parameters are of great importance for the correct characterization of laser beams. Edge steepness and plateau uniformity are the most significant parameters of beam distributions with a nearly flat-top profile [33, 34, 48].

The first quality parameter on the beam intensity distribution is the edge steepness, and is defined as:

$$S_{10\%,90\%} = \frac{A_{10\%} - A_{90\%}}{A_{10\%}}, \tag{17}$$

where $A_{10\%}$ and $A_{90\%}$ are the effective intensity areas with intensity values $I_{10\%}$ and $I_{90\%}$, respectively. Please note that the intensity values (I) are equivalent to I_{FTB} and are valid for the remaining part of the manuscript. The $S_{10\%,90\%}$ value approaches 0, which means that the edges of the distribution become steeper. The second parameter is the plateau uniformity of the beam, which defines the uniform flat profile ability of a beam on its intensity distribution [48]. The plateau uniformity for continuous wave lasers can be expressed as:

$$U_P = \frac{\Delta I_{FWHM}}{I_{max}}, \tag{18}$$

where ΔI_{FWHM} is the full width at half maximum value of the intensity distribution, and I_{max} is the maximum intensity. when $U_P \rightarrow 1$, and the flat-top deviates from the perfect flat top.

2.5.2 RMSE Calculation

One of the best methods of measuring the efficiency of beam reconstruction using an IFTA is the RMSE calculation. In the Fourier plane, which is located in the focal length f, depending on the reconstructed intensity, the accuracy is defined as the root-mean-square error

$$\text{RMSE} = \frac{1}{\sqrt{N}} \sqrt{\sum_{i,j} (I_p(i,j) - I_{\text{target}}(i,j))^2} \tag{19}$$

where I_{target} is the target intensity distribution,I_p is the reconstructed intensity distribution, i and j are the pixel indices, and N is the number of pixels in the non-zero target region. I_{target} can be generated as a perfect square FTB using Eqs. (7) and (8), while I_p, the far-field reconstructed intensity distributions of the square FTBs, are obtained based on Eqs. (7) to (13).

3 Simulation test of IFTAs

A numerical simulation to test the effectiveness of the MMRAF algorithm is presented in this section. As an input to Eq. (16), a laser beam with a large diameter of 20 mm with an assumed uniform plane wave distribution was used to illuminate the SLM device. This uniform plane wave was implemented experimentally using a collimation unit, as explained in the next section. After 75 iterations of Eq. (6) for the MRAF and MMRAF algorithms with $m = 0.3$, the FTB patterns of the intensity distribution were reconstructed in the far field. For the MMRAF algorithm, $\Delta = 13.31$ µm was computed by Eq. (10) (where $N = 1920$ pixels (horizontal)) with a lens of focal length 250 mm. A CMOS camera was placed at a distance equal to the focal length to record the optical reconstructed patterns. In the MMRAF algorithm, we used Eqs. (7) to (13) to calculate the reconstructed FTB intensities, and we assumed that all windows of the SLM are the SR' (i.e., 1920×1080), which gives the maximum possible source size of the beam. Thus, we can create this window inside the algorithm as the SLM window and work on it by taking some pixels as 100×100, 164×164, etc. These generated pixels give the required source sizes (i.e., $\propto_{x,y}$). For the classical MRAF algorithm, however, the target or the source size that enters the algorithm has a fixed number of pixels, which changes every time to obtain another source size, and does not depend on the total number of pixels in the SLM window. The theoretical calculations in the study for the MMRAF and MRAF algorithms were performed based on the above assumptions.

4 Results and discussions

4.1 Theoretical comparison of MRAF and MMRAF

Based on Eq. (6) for the MRAF and MMRAF algorithms, the final kinoform holograms were calculated, and a comparison between the reconstructed intensity distributions of FTBs is presented. Please note that for a better assessment, all generated and observed intensity values are normalized with respect to their own maximum values. Figure 3 shows the results of the 3D reconstructed square FTB intensities at the far field using MRAF and MMRAF algorithms for beam sizes of $\alpha_{x,y} = 1.66$ mm. From Fig. 3, the intensity occupied by the FTBs with the MMRAF algorithm in the far field is higher than that of the FTBs using the MRAF algorithm. This enhancement in beam intensity reconstruction of the FTBs is based on the advantage of the MMRAF algorithm to find the best phase modulation at the SLM plane. The corresponding profile plots of the beam intensities of the square FTBs against the spatial position based on Fig. 2 are also shown in Fig. 4. As can be seen in Fig. 4, the intensity of the far-field distributions of the square FTBs with both the MMRAF and MRAF algorithms have speckles in the profiles that are an actual result from the IFTAs. From Fig. 4,

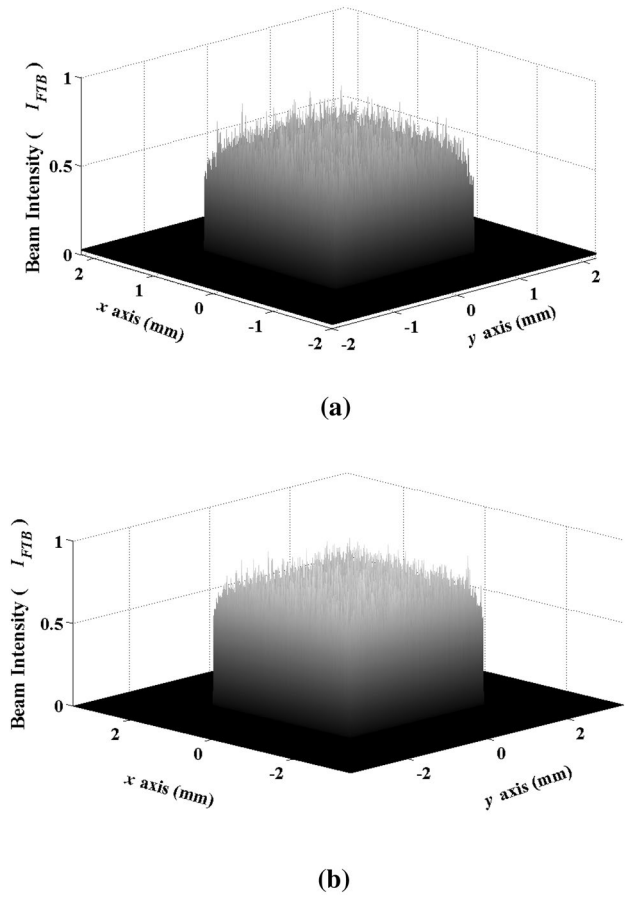

Fig. 3 3D plot of beam intensities for square FTB at $\propto_{x,y}$ = 1.66 mm using **a** MRAF and **b** MMRAF algorithms

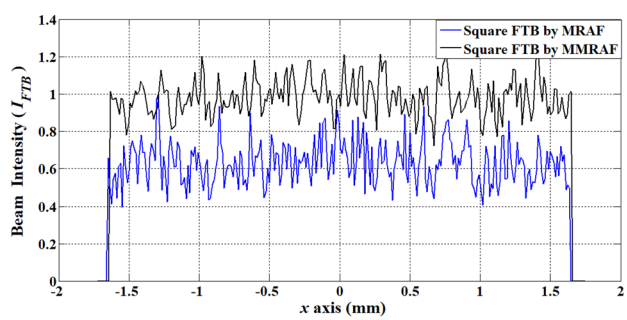

Fig. 4 Profile plots of beam intensities of square FTB for MRAF and MMRAF algorithms at $\propto_{x,y}$ =1.66 mm

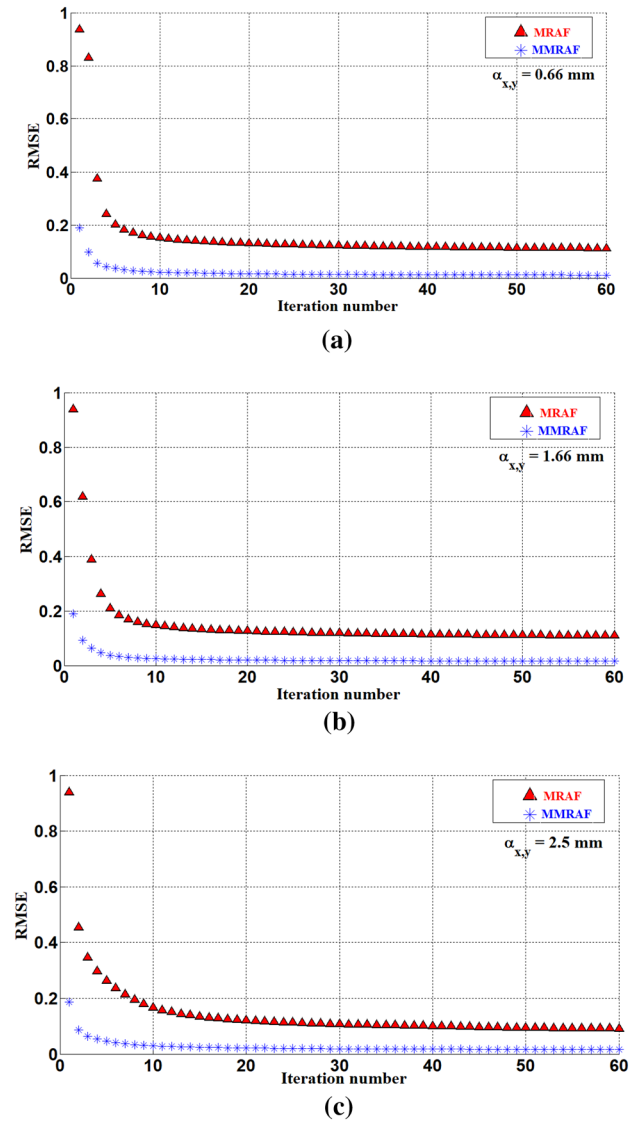

Fig. 5 RMSE as a function of iteration number for MRAF and MMRAF algorithms for beam sizes: **a** $\propto_{x,y}$ = 0.66 mm, **b** $\propto_{x,y}$ =1.66 mm, and **c** $\propto_{x,y}$ = 2.5 mm

we notice that the edge steepness and uniformity of the intensity oscillation of the square FTB seem better in the MMRAF algorithm.

Although the intensity evaluation of algorithms gives a perception about their beam reconstruction capability, the golden standard [28, 29] is used to measure the performance of a beam reconstruction based on the RMSE analysis. Thus, the beam reconstruction efficiency of the IFTA algorithms was evaluated based on the RMSE calculation.

To show the improvement of the MMRAF method over the MRAF, using the target and reconstructed beam intensities, we calculated RMSEs based on Eq. (19). The RMSE comparison of the MRAF and MMRAF algorithms as a function of the iteration number for selected beam sizes is shown in Fig. 5. As shown in Fig. 5 for all beam sizes, the RMSE decays exponentially with increasing number of iterations for both the MRAF and MMRAF algorithms. Most real-time beam shaping systems in laser processing require fast and better reconstructed intensities. This means that low RMSE values with small iteration numbers are necessary. Based on these criteria, the assessment of the figure shows

that for all source sizes, the MMRAF algorithm performs better compared with the MRAF algorithm for changing iteration numbers. It can be seen that the MMRAF has a fast convergence of target intensities for square FTBs with fewer iteration numbers. Even with changing beam sizes, it still provides a fast and efficient reconstruction. For all beam sizes using the MMRAF algorithm, the theoretical performance of the beam intensity shaping is improved to less than 0.02 RMSE. This rate is achieved by 24 iterations of beam sizes of $\propto_{x,y}$ = 2.5 mm, 19 iterations of $\propto_{x,y}$ = 1.66 mm and 12 iterations of $\propto_{x,y}$ = 0.66 mm which are very small and comparable to other proposed methods [26, 27]. Although the beam sizes change, the MMRAF still yields efficient reconstructed beam intensities at the detector plane with a minor RMSE.

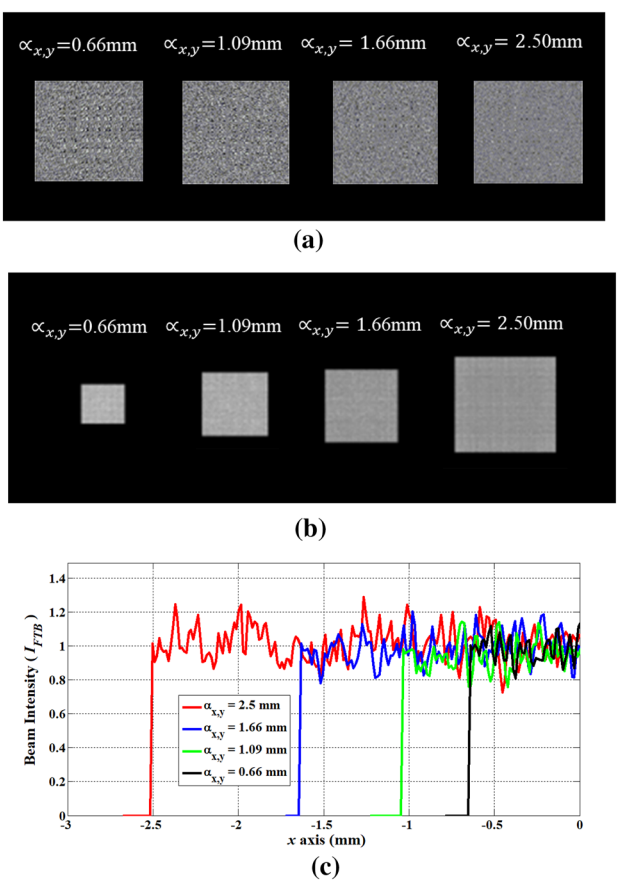

Fig. 6 **a** Phase kinoform holograms of square FTBs using MMRAF algorithm for different beam sizes **b** Theoretical reconstructed intensity distributions of square FTBs for different beam sizes **c** Corresponding Profile plots of beam intensities

4.2 Theoretical generated dynamic FTBs with the MMRAF method

Based on Eqs. (7) and (8), different numbers of pixels as the *SR′*, such as 100 × 100, 164 × 164, 250 × 250, and 376 × 376 are formed to generate varying beam sizes of square FTBs as $\propto_{x,y}$ = 0.665 mm, 1.09 mm, 1.66 mm, and 2.5 mm symmetrically in the *x* and *y* directions. The results of phase kinoform holograms for square dynamic FTBs with different source sizes are displayed in Fig. 6(a). The beam sizes were computed using Eqs. (9) and (10). The far-field reconstructed intensity distributions of the square FTBs with different beam sizes were obtained as grayscale images based on Eqs. (7) to (13) for the MMRAF algorithm. The results of the far-field intensity distributions for controllable beam sizes are shown in Fig. 6(b). We used the same setting of beam sizes for the remaining part of the study for both theoretical calculations and experimental analyses of the square FTBs.

In this manuscript we use two different parameters for analyzing the beam reconstruction capability of the MMRAF method in terms of uniformity; one is plateau uniformity and the other is image homogeneity. Each of them gives a different but supplementary perspective of the beam quality measure. To make a reliable comparison of these figures, we used homogeneity analysis on the generated square FTBs. Homogeneity (*H*) is known as the inverse difference moment and is used to determine the homogeneity of a digital image based on intensity distributions along pixels. Using Eq. (21) of Reference [49] homogeneities of the FTBs for $\propto_{x,y}$ = 0.665 mm, 1.09 mm, 1.66 mm, and 2.5 mm can be calculated as *H* = 0.9451, 0.9556, 0.9636, and 0.9738, respectively. Homogeneity values close to 1 are assumed to be highly homogeneous; therefore, according to these data with beam size increments, the homogeneity of FTBs improves with beam size. In addition, comparing the *H* data for individual beam sizes, it can be seen that they are very close to each other. This means that with increasing beam sizes, FTB reconstruction performs small oscillations along the *H* values, which is a supportive sign of the dynamic beam shaping efficiency of the MMRAF algorithm. Because large beam sizes (such as 2.5 mm) require more pixels as the *SR′* for beam reconstruction, FTBs exhibit more homogenous intensity distributions for those beam sizes. The corresponding profile plots of the beam intensities of the square FTBs against the spatial position with variation in beam size based on Fig. 6(b) are shown in Fig. 6(c).

As shown in Fig. 6(c), the intensity area of the square FTBs increases with increasing beam size. Even when changing the shape profile of the FTBs using the MMRAF algorithm, we still obtained a well-reconstructed beam intensity with no significant oscillation constrained to each shape

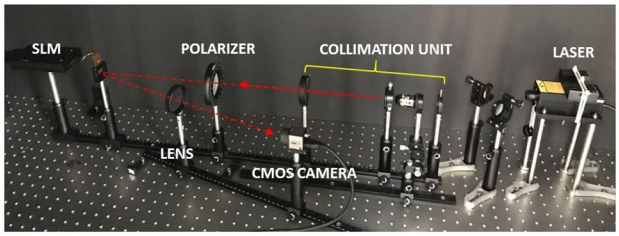

Fig. 7 Experimental setup

profile. Thus, with the MMRAF algorithm, dynamic control of beam shaping is possible based on varying SR' pixels. As can be seen, with increasing beam sizes, the edge steepness of the intensities maintains its steep shape consistently for different spatial positions.

4.3 Experimental generated dynamic FTBs with the MMRAF method

4.3.1 Experimental setup

The generation of square FTBs using the MMRAF algorithm was experimentally performed through the use of a phase-only SLM. A laser (300mW, CHANGCHUN OPT: MDL-III852R) beam emitting a continuous wave Gaussian beam of wavelength 852 nm was used. The Gaussian beam was converted into a plane wave front to provide uniform illumination of the SLM active area using a specific collimation unit (~ 20X magnification). The collimation unit also ensured efficient modulation through the SLM pixels and consisted of an objective lens (Olmypus, UPLFLN10X2), a pinhole (Thorlabs, P100S), and a biconvex lens (Thorlabs, LB1607-B) with a focal length of 175 mm. To control the phase modulation of the light field incident on the SLM, a polarizer (Thorlabs, LPNIRE200B) was used. In this study, the SLM device used was a Holoeye-Pluto NIR II based on reflective LCOS micro displays of 1920 × 1080 pixel resolution with a small pixel pitch (Δ') of 8 µm and a reflectivity of 62% for the wavelength used. The plane of the SLM was Fourier transformed with a biconvex lens of focal length $f = 250$ mm. The far-field intensity distributions were recorded as grayscale images using a digital CMOS camera (Thorlabs, DCC3240N-NIR) at 1280 × 1024-pixel resolution and a pixel pitch of 5.3 µm, which was placed and fixed at the output plane (detector plane), as shown in Fig. 7. The encoded lens in Eq. 5 as a digital focusing executed by the SLM is required to concentrate the light while the lens in Fig. 7 is used for the far-field distributions.

Experiments were conducted to compare the optical performances of the controls on the number of pixels being used in the generation of the required beam sizes, and the far-field intensity distributions of the controllable square FTBs with

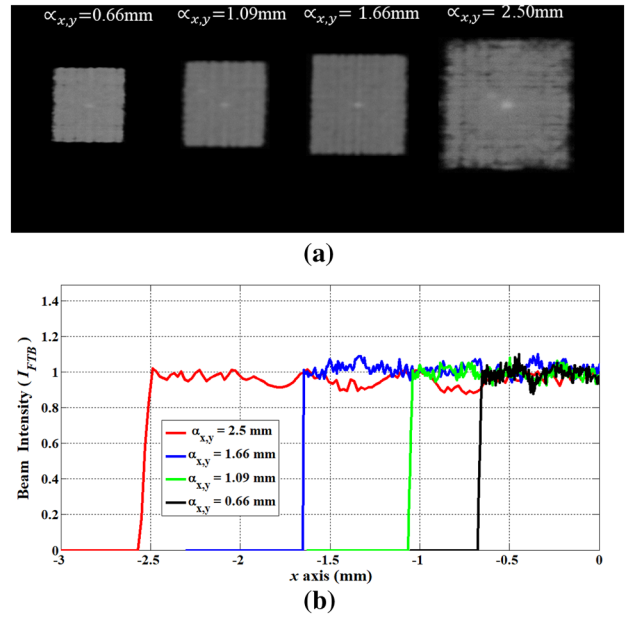

Fig. 8 **a** Experimental recorded intensity distributions of square FTBs for different beam sizes using the MMRAF algorithm. **b** Profile plots of recorded beam intensities

the MMRAF algorithm. Based on the optical setup that provides the required flat wavefront distribution, the far-field intensity distributions of the controllable dynamic FTBs with the MMRAF algorithm were recorded at the output plane using a digital CMOS camera. The beam sizes of the controllable FTBs were measured based on the following formula: $MSS = N_{CCD} \times \Delta_{CCD}$ where MSS is the measured beam size, N_{CCD} is the number of pixels on the camera, and $\Delta_{CCD} = 5.3$ µm is the pixel pitch of the camera. Using the MMRAF algorithm based on Eqs. (7), (8), and (13), final kinoform holograms were generated for different beam sizes of FTBs (based on corresponding pixels of the SR' of 100 × 100, 164 × 164, 250 × 250, and 376 × 376). These were loaded into the SLM device to perform the experiment.

4.3.2 Experimental results

The recorded far-field intensity distributions of the square FTBs by the CMOS camera located at the back focal plane of the Fourier lens are shown in Fig. 8(a). The widths of these FTBs were measured as $\propto_{x,y} = 0.665$ mm, 1.09 mm, 1.66 mm, and 2.5 mm.to determine the beam sizes of the FTBs.

To make a better assessment of these figures, homogeneity analysis was performed on the recorded square FTBs, similar to the theoretical results. Using Eq. (21) of Reference [48], homogeneities of the recorded FTBs for $\propto_{x,y} =$ 0.665 mm, 1.09 mm, 1.66 mm, and 2.5 mm were calculated as $H = 0.9425$, 0.9508, 0.9591, and 0.9309, respectively.

With increasing beam size, the homogeneity of the FTBs increased. Because the H data for individual beam sizes are very close to each other, we can conclude that the FTBs reconstruction produced adequate beam shaping with varying beam sizes. The corresponding profile plots of the recorded beam intensities of the square FTBs with variations in beam size based on Fig. 8(a) are shown in Fig. 8(b). As shown in Fig. 8(b), the intensity area of the square FTBs increases with increasing beam size. In addition, with increasing beam size, the edge steepness of the intensities steadily maintains its steep shape through spatial extension. With the aid of the MMRAF algorithm, although the beam size of the FTBs changes, we still obtained well-recorded beam intensities with small intensity oscillations. As can be seen from these results, the dynamic beam shaping method gives promising beam generation and controlling capability at the Fourier plane based on SR' variations. From Fig. 8(a) and (b), low noise with a clean background is observed through the recorded intensity patterns owing to the effective reconstruction capability of the MMRAF algorithm, which reduces the effect of the zero intensity value in the Fourier plane. It can be concluded from Fig. 8(a) and (b) that the recorded far-field intensity distributions of the dynamic FTBs with increasing number of pixels defining the beam sizes become more homogeneous and uniform. However, when the number of pixels defining the width of the required beam profile reaches the full size of the desired shape inserted into the Fourier plane, the recorded far-field intensity distributions of the dynamic FTBs are distorted because the unmodulated light increases owing to the increase in the inactive area of the SLM. Because the modulation efficiency is degraded, the intensity distribution of the edge of the beam size of $\propto_{x,y}$ = 2.5mm is slightly corrupted.

Based on the theoretical and experimental results obtained, the success of the MMRAF algorithm for dynamic beam shaping can be evaluated. With the MMRAF algorithm, the far-field intensity distributions of the square FTBs can be obtained by controlling the number of pixels inside the non-zero SR' used in the determination of the phase holograms for different structures of the FTB distributions. In this scope, the theoretical and experimental 3D far-field intensity distributions of the square FTBs for a selected beam size ($\propto_{x,y}$= 1.66 mm), are shown in Fig. 9. The corresponding profile plots of the experimental and theoretical beam intensities of the square FTB are given in Fig. 10. It can be concluded from Figs. 9 and 10 that the experimental and theoretical results of FTBs obtained by the MMRAF algorithm are very close to each other in terms of intensity distributions along the spatial positions and their edge steepness. From the standpoint of homogeneity, H is calculated as 0.9636 and 0.9591 for the theoretical and experimental results, respectively, which are very similar. Furthermore, with increasing beam size ($\propto_{x,y}$) homogeneities of the reconstructed beams for both theoretical and experimental results agree with each other. When changing the beam profile of the square FTBs using the MMRAF algorithm, we obtained the same characteristics as the FTBs without loss of any information to define it. This comparison verifies that the MMRAF algorithm is an effective method for generating a different beam profile structures with controllable beam sizes and intensities.

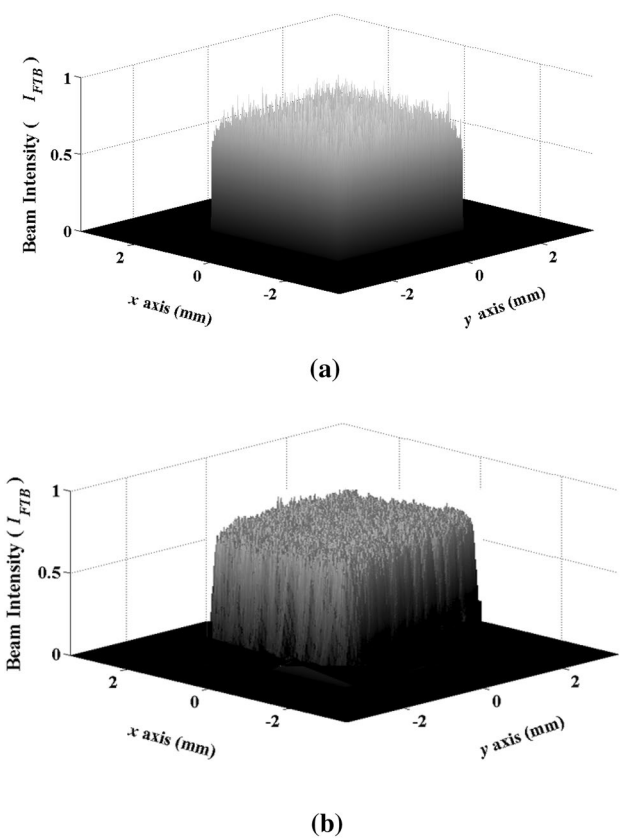

Fig. 9 Intensities of the square FTBs by the MMRAF algorithm at $\propto_{x,y}$ = 1.66 mm: **a** theoretical, and **b** experimental

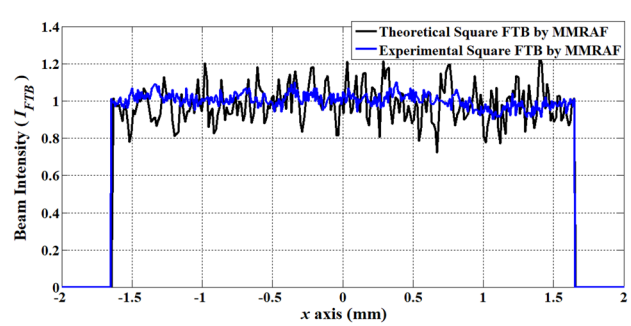

Fig. 10 Profile plots of theoretical and experimental beam intensities of square FTBs for MMRAF algorithm at $\propto_{x,y}$ = 1.66 mm

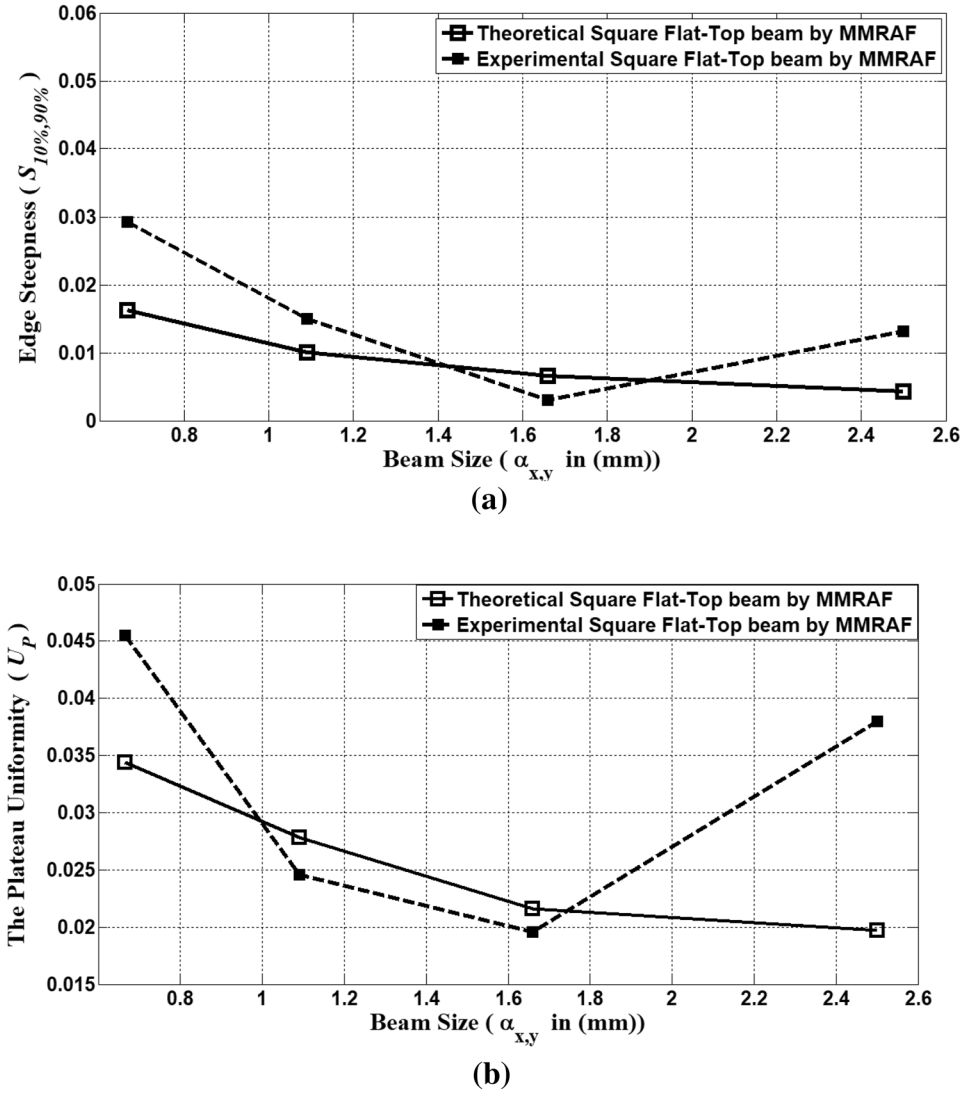

Fig. 11 **a** Theoretical and experimental edge steepness comparison of square FTBs using the MMRAF algorithm. **b** Theoretical and experimental plateau uniformity comparison of square FTBs using the MMRAF algorithm

Although the intensity evaluation of the MMRAF algorithm provides insight into their beam reconstruction capability, we also conducted fineness degree analyses of the dynamic FTBs in terms of edge steepness and plateau uniformity.

4.4 Edge steepness and plateau uniformity of dynamic FTBs using the MMRAF algorithm

In this section, edge steepness and plateau uniformity are evaluated as quality criteria for beam distribution with a nearly flat-top profile to demonstrate the efficiency of our proposed method. Based on Eqs. (17) and (18), the theoretical and experimental results of edge steepness and plateau uniformity of the square FTBs obtained by the MMRAF algorithm for different beam sizes ($\propto_{x,y}$) are illustrated in Fig. 11. From Fig. 11(a), it seems that the theoretical and experimental values of edge steepness decrease with increasing beam size of square FTBs. From Fig. 11(a), it can be seen that the generated dynamic square FTBs have edge steepness values that approach zero, which means that our generated beams are very close to the structure of a perfect FTB.

It can be seen that the edge steepness values of both theoretical calculations and experimental measurements are consistent with each other. The lower edge steepness values indicate a steeper profile of the flat beam intensity distribution. For all measurements and calculations, it can be seen that both the experimental and theoretical edge steepness values are below 0.03, which means higher vertical edge laser beams. These results make the MMRAF algorithm reconstruction method a very efficient flat beam generation technique, especially for laser machining applications requiring highly steep fabrication of complex patterns. In terms of dynamic beam shaping analyses of our method, we find that even if the beam size increases, the edge steepness variation

is within the limits of 0.005 and 0.03. Therefore, there is no significant variation or loss of information through the uniformity of the reconstructed beam. From Fig. 11(b), the theoretical results of the dynamic square FTBs show that the plateau uniformity decreases when the beam size ($\propto_{x,y}$) increases. The experimental plateau uniformity results of the FTBs exhibit the same general decreasing trend with increasing beam size. The small uniformity values of the FTBs provide flatter intensity shapes. While the uniformity of the theoretical values fluctuates between 0.034 and 0.02, the experimental values of plateau uniformity are in the region between 0.046 and 0.02. When the beam sizes increase, the plateau uniformity of the generated square FTBs is enhanced to achieve more uniform FTB profiles. The square FTB had satisfactory lower values of plateau uniformity for all beam sizes. In general, the theoretical and experimental uniformity results agree with each other except for the beam size of $\propto_{x,y} = 2.5$mm. The reason may originate from the fill factor of the SLM, which is approximately 93%, so there can be unmodulated light in the experimental reconstruction process of FTBs. Because larger beams require more pixels for phase reconstruction, modulation will be highly coupled to unmodulated light as a part of the diffraction effects due to the increasing inactive area of the SLM. Thus, the plateau uniformity of the beam at the detector plane may be degraded. To assess the ability of dynamic beam shaping based on the MMRAF algorithm, we can analyze Fig. 11(b) as well. With increasing beam size, the variation in plateau uniformity is small and consistent within the limits of 0.02–0.046. However, the variation in the experimental flat beam uniformity is higher and has a larger interval with changing beam size. Although there is a small difference between the theoretical and experimental results, it is seen that uniformity remains stable with increasing beam size in terms of beam shaping capacity. Making analyses of Homogeneity and the plateau uniformity results, it is seen that with increasing beam sizes Homogeneity improves, and a similar improvement of plateau uniformity trend can be seen in Fig. 11(b) as well. So these results are consistent with each other we can conclude that Homogeneity and plateau uniformity is two different but a complementary measure of the Flat beam intensity uniformity.

5 Conclusions

In this study, a new method is proposed to generate dynamic FTBs based on the calculation of the phase kinoform hologram of the desired intensity distribution at the far-field plane. This new methodology uses the variation in the number of pixels that describe the width of the signal region of the MRAF algorithm to obtain FTBs with controllable beam size and intensity. A detailed theoretical and experimental comparison was performed with the MMRAF algorithm to implement a kinoform hologram for the dynamic control of square FTB intensity distributions. This comparison was made using standard parameters, such as edge steepness, plateau uniformity, and the intensity of the FTBs. It was found that the MMRAF algorithm provided high intensity uniformity in the plateau region and formed steep edged beams at the detector plane. Furthermore, the improvement of our proposed method was compared with the classical MRAF algorithm based on RMSE calculations. It was found that the MMRAF algorithm showed better performance when compared with the MRAF algorithm for changing the iteration numbers. Using our proposed method, the performance of beam intensity shaping is improved to less than 0.02 RMSE with minimum iterations of phase reconstruction. The far-field patterns of the square FTBs obtained by the MMRAF algorithm have a clean background for the reconstructed far-field distributions. Homogeneity analyses of reconstructed beams verified that although the beam size of the FTBs changed, we still obtained well-recorded beam intensities with uniform and flat-topped intensities.

Since the proposed dynamic beam shaping method depends on the determination of the size of SR' region, a flexible Fourier plane reconstruction capability is provided. With this distinct ability, the MMRAF method can be used to generate other kinds of flat beams such as circular flat-topped or elliptical flat-topped beams. We conclude that the MMRAF algorithm can be used to generate any desired profile of the FTB with a controllable beam size based on the phase-only SLM without loss of flatness of the FTB. Moreover, we proved that the MRAF algorithm can be used as a power tool for FTBs, regardless of the shape of the target to be generated or inserted into the Fourier plane. The experimental and theoretical results for all cases with different beam sizes obtained by the MRAF algorithm are compatible. Thus, we introduced a flexible tool for FTBs used in fields that require a flattened irradiance profile with a uniform central region, which makes it a faultless tool for many applications.

Acknowledgements The authors acknowledge the support provided by Çankaya University and the Department of Laser and Optoelectronics Engineering at Nahrain University.

References

1. D. Shealy, Proc. of SPIE **4770**, 29–46 (2002)
2. F. Dickey, L. Weichman, R. Shagam, Proc. of SPIE **4065**, 338–348 (2000)
3. D. Shealy, J. Hoffnagle, Appl. Opt. **45**, 5118–5131 (2006)
4. B. Frieden, Appl. Opt. **4**, 1400–1403 (1965)
5. C. Ih, Appl. Opt. **11**, 694–695 (1972)
6. W. Simmons, G. Leppelmeier, B. Johnson, Appl. Opt. **13**, 1629–1632 (1974)

7. C. Han, Y. Ishii, K. Murata, Appl. Opt. **22**, 3644–3647 (1983)
8. M. Karim, A. Hanafi, F. Hussain, S. Mustafa, Z. Samberid, N. Zain, Opt. Lett. **10**, 470–471 (1985)
9. F. Dickey, B. O'Neil, Opt. Eng. **27**, 999–1007 (1988)
10. V. Kermene, A. Saviot, M. Vampouille, B. Colombeau, C. Froehly, T. Dohnalik, Opt. Lett. **17**, 859–861 (1992)
11. K. Nemoto, T. Fujii, N. Goto, T. Nayuki, Y. Kanai, Opt. Lett. **21**, 168–170 (1996)
12. K. Nemoto, T. Fujii, N. Goto, H. Takino, T. Kobayashi, N. Shibata, K. Yamamura, Y. Mori, Appl. Opt. **36**, 551–557 (1997)
13. J. Pu, H. Zhang, Appl. Opt. **37**, 4200–4205 (1998)
14. S. Zhang, G. Neil, M. Shinn, Opt. Express. **11**, 1942–1948 (2003)
15. D. Shealy, S. Chao: Proc. of SPIE 5525 (2004).
16. R. Gerchberg, W.O. Saxton, Optik **35**, 237–246 (1972)
17. F. Li, Z. Lu, H. Li, J. Liao, Proc. of SPIE **4095**, 189–195 (2000)
18. Y. Zhao, Y. Li, Q. Zhou, Proc. of SPIE **5175**, 77–87 (2003)
19. G. Račiukaitis, E. Stankevičius, P. Gečys, M. Gedvilas, C. Bischoff, E. Jäger, U. Umhofer, F. Völklein, J. Laser Micro/Nanoeng. **6**, 37–43 (2011)
20. D. Wang, X. Liu, H. Wang, J. Zhang, L. Wu, Proc. of SPIE **8759**, 1–6 (2013)
21. M. Pasienski, B. DeMarco, Opt. Express. **16**, 2176–2190 (2008)
22. T. Harte, G. Bruce, J. Keeling, D. Cassettari, Opt. Express. **22**, 26548–26558 (2014)
23. H. Ma, Z. Liu, P. Zhou, X. Wang, Y. Ma, X. Xu, J. Opt. A Pure Appl. Opt. **12**, 045704 (2010)
24. A. Hendriks, D. Naidoo, F. Roux, C. López-Mariscal, A. Forbes, Proc. of SPIE **8490**, 1–8 (2012)
25. D. Wang, B. Jin, Y. Wang, P. Jia, D. Cai, Y. Gao, IEEE Photonics J. **8**, 1–7 (2016)
26. J.M. Maxson, A.C. Bartnik, I.V. Bazarov, Appl. Phys. Lett. **105**, 171109 (2014)
27. J. Liang, R.N. Kohn Jr., M.F. Becker, D.J. Heinzen, Appl. Opt. **49**, 1323–1330 (2010)
28. H. Pang, W. Liu, A. Cao, Q. Deng, Opt. Commun. **433**, 44–51 (2019)
29. T. Häfner, J. Heberle, D. Holder, M. Schmidt, J. Laser Appl. **29**, 022205 (2017)
30. C. Chang, J. Xia, L. Yang, W. Lei, Z. Yang, J. Chen, Appl. Opt. **54**, 6994–7001 (2015)
31. A. Jesacher, C. Maurer, S. Schwaighofer, M. Bernet, M. Ritsch-Marte, Opt. Express. **16**, 4479–4486 (2008)
32. J. Shi, D. Wei, C. Hu, M. Chen, K. Liu, J. Luo, X. Zhang, Opt. Express. **29**, 7084–7099 (2021)
33. S. Bollanti, P. Di Lazzaro, D. Murra, E. Tefouet Kana, G. Felici: Appl. Phys. B Lasers Opt. 78, 195–198 (2004)
34. EN ISO 13694 "Test Methods for Laser Beam Power (Energy) Density Distribution" (2000).
35. S. Jacobsson, A. Lundgren, J. Johansson, Int. J. Infrared Millimeter Waves. **11**, 1251–1261 (1990)
36. Z. Zalevsky, D. Mendlovic, R.G. Dorsch, Opt. Lett. **21**, 842–844 (1996)
37. B. Kress, P. Meyrueis, Digital diffractive optics: An introduction to planer diffractive optics and related technology: John Wiley & Sons Ltd, England, (2000).
38. S. Jacobsson, S. Hard, A. Bolle, Appl. Opt. **26**, 2773–2781 (1987)
39. C. Schulze, D. Flamm, M. Duparré, A. Forbes, Opt. Lett. **37**, 4687–4689 (2012)
40. C. Rosales-Guzmán, A. Forbes, How to shape light with spatial light modulators: Press of SPIE, USA, (2017).
41. C. Zhang, Y. Hu, W. Du, P. Wu, S. Rao, Z. Cai, Z. Lao, B. Xu, J. Ni, J. Li, G. Zhao, D. Wu, J. Chu, K. Sugioka, Sci. Rep. **6**, 1–9 (2016)
42. H.T. Eyyuboğlu, Ç. Arpali, Y. Baykal, Opt. Express. **14**, 4196–4207 (2006)
43. P. Senthilkumaran, F. Wyrowski, H. Schimmel, Opt. Lasers Eng. **43**, 43–56 (2005)
44. S. Reichelt, N. Leister, J. Phys. Conf. Ser. **415**, 1–10 (2013)
45. H. Zhang, J. Xie, J. Liu, Y. Wang, Appl. Opt. **48**, 5834–5841 (2009)
46. B. Saleh, M. Teich, *Fundamentals of Photonics*, 2nd edn. (John Wiley & Sons Inc., New York, 2007)
47. J. Goodman, Introduction to Fourier Optics, 2nd ed.: McGraw-Hill Companies, New York. (1998)
48. J.J. Yang, M.R. Wang, Opt. Eng. **42**, 3106–3113 (2003)
49. D.Y. Alsaka, Ç. Arpali, S.A. Arpali, Opt. Rev. **25**, 625–637 (2018)

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.